Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1 – INSTRUCTION SCREEN

Description

This screen introduces the user to the memory recall task in a **clear, structured, and visually calm layout**. The design uses **phase-based chunking** to explain the process without overwhelming the user.

UI Elements

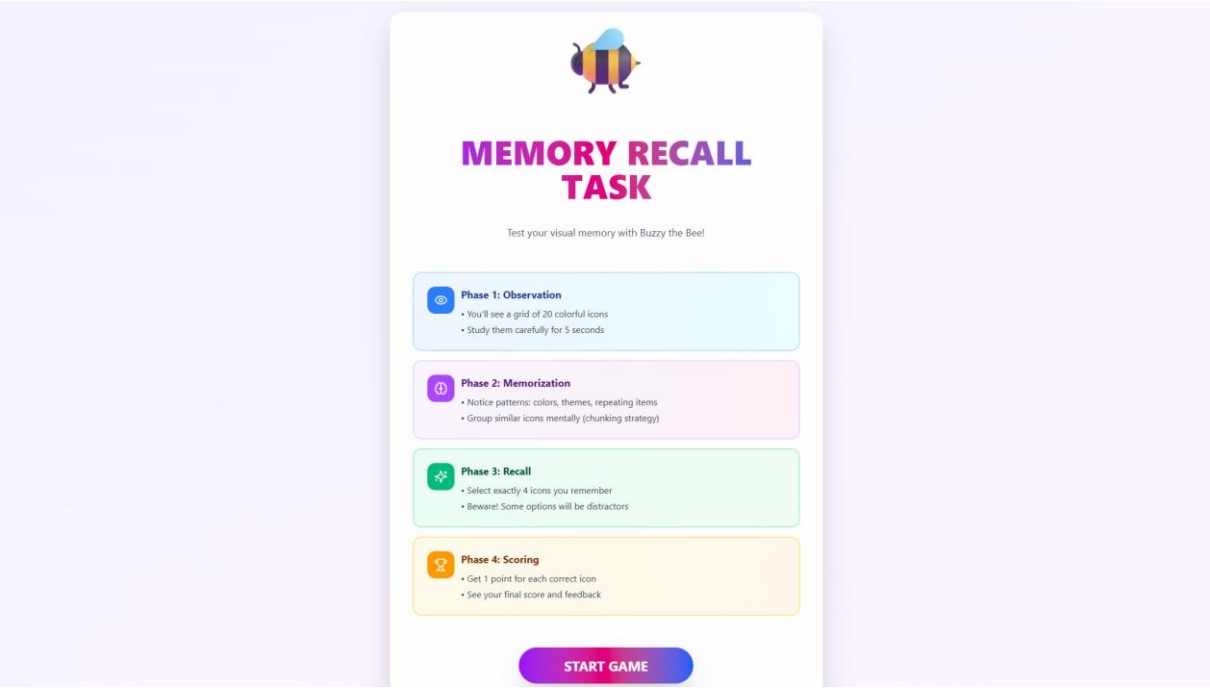
- Bee mascot at the top for friendly visual identity
- Title: **"MEMORY RECALL TASK"**
- Subtitle: *"Test your visual memory with Buzzy the Bee!"*
- Four instruction cards:
  - Phase 1: Observation
  - Phase 2: Memorization
  - Phase 3: Recall
  - Phase 4: Scoring
- Primary CTA button: **START GAME**

Chunking Analysis

- Instructions are divided into **four clearly labeled phases**, making the task easier to understand.
- Each phase card contains **only two short bullet points**, reducing cognitive load.
- Color-coded cards visually separate each phase, supporting faster scanning and comprehension.
- Large spacing, soft shadows, and minimal text enhance readability and focus.

Cognitive Benefit

Chunking the instructions into phases helps users form a **mental model of the task flow** before starting, improving task confidence and comprehension.



FRAME 2 – OBSERVATION / CHUNKING PHASE

Description

This screen represents the **visual encoding stage**, where users observe and memorize icons within a limited time.

UI Elements

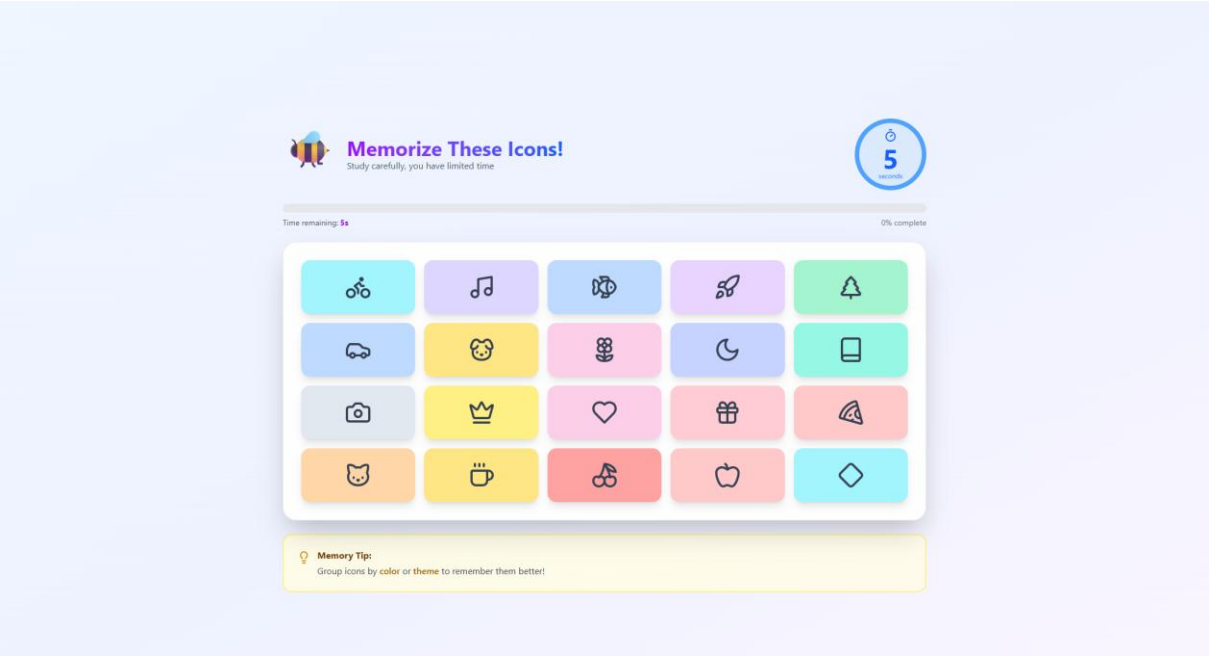
- Header: **"Memorize These Icons!"**
- Subtext: *"Study carefully, you have limited time"*
- Circular countdown timer showing **5 seconds**
- Horizontal progress bar indicating time remaining
- Grid of **20 colorful icons** arranged in a structured layout
- Memory tip card at the bottom:
    - *"Group icons by color or theme to remember them better!"*

Chunking Strategy

- Icons are grouped using:
    - Similar colors
    - Familiar categories (objects, animals, symbols)
    - Repeating visual patterns
- Rounded cards and consistent spacing help users identify groups quickly.

Cognitive Benefit

Users rely on **pattern recognition and grouping** rather than memorizing individual icons, strengthening short-term memory through chunking.



FRAME 3 – RECALL / SELECTION PHASE

Description

This screen tests the user's ability to **retrieve previously seen information** by selecting remembered icons.

UI Elements

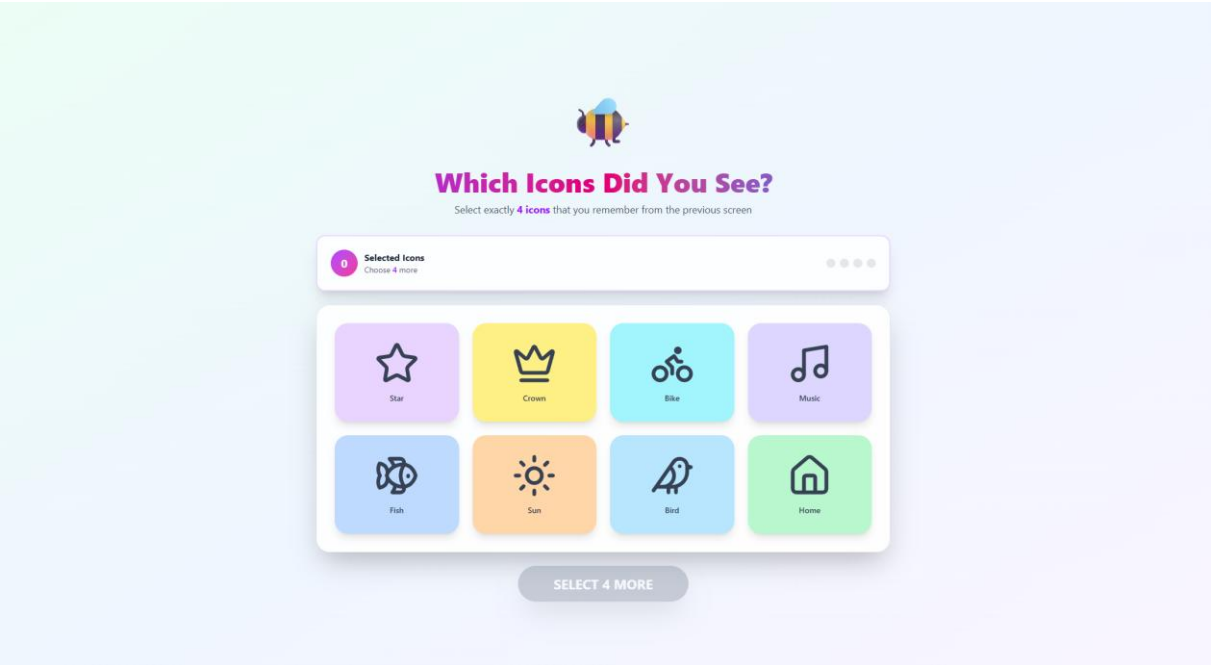
- Title: **"Which Icons Did You See?"**
- Instruction: *"Select exactly 4 icons that you remember from the previous screen"*
- Selection counter:
  - Shows how many icons are selected and how many remain
- Grid of **8 icon options**, including:
  - Icons from the observation phase
  - Distractor icons not shown earlier
- Disabled CTA button until 4 icons are selected

Interaction Design

- Selected icons are visually highlighted
- The selection counter provides real-time feedback
- Clear constraints prevent over-selection

Cognitive Benefit

The presence of distractors evaluates **memory accuracy**, while clear feedback prevents guessing and encourages deliberate recall.



FRAME 4 – RESULTS / FEEDBACK SCREEN

Description

This screen provides **instant performance feedback** in a positive, encouraging manner.

UI Elements

- Bee mascot at the top
- Success message: **"EXCELLENT!"**
- Score card displaying:
  - Correct answers (e.g., **3 / 4**)
  - Accuracy percentage (e.g., **75% Accuracy**)
- Feedback message reinforcing motivation
- Action buttons:
  - **Play Again**
  - **New Game**
  - **Exit**
- Memory enhancement tips section at the bottom

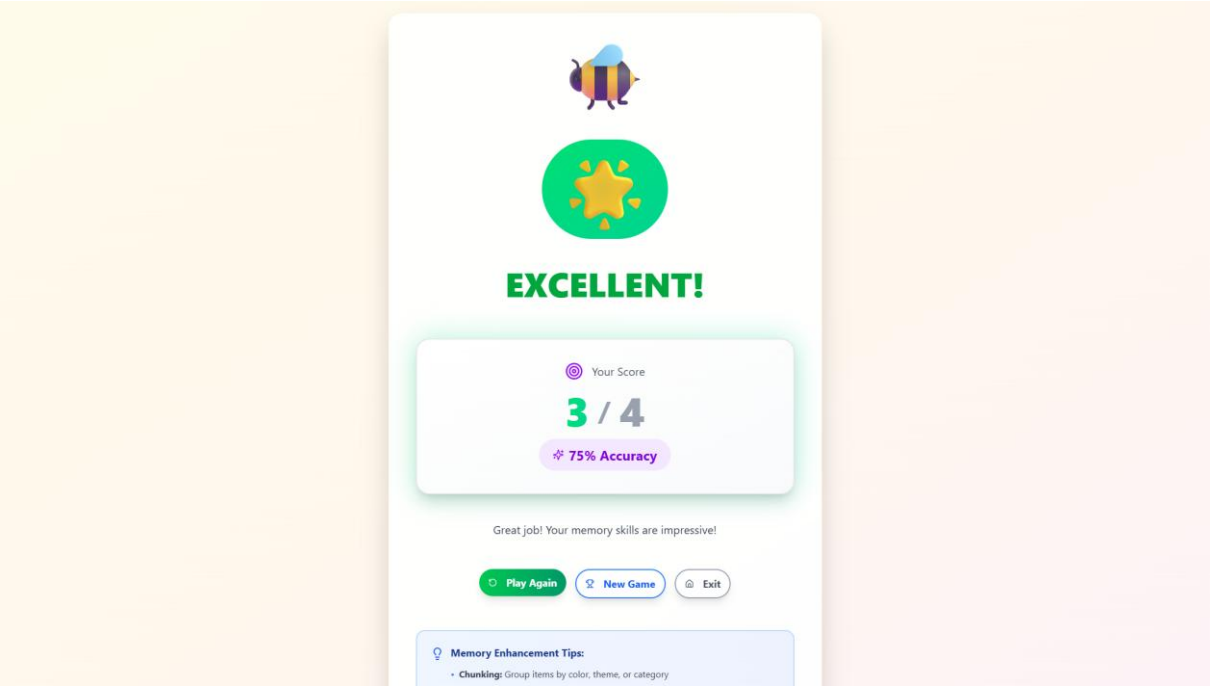
Feedback Design

- Large typography emphasizes results
- Soft glow and rounded card elevate focus on the score

- Action buttons give users control over next steps

Cognitive & UX Benefit

Immediate feedback reinforces learning, while replay options encourage **repetition and improvement**, strengthening memory retention over time.



PROTOTYPE LINK:

https://finite-silo-13106053.figma.site/results